class10: Structural Bioinformatics (pt. 1)

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PDB statistics

The Protein Data Bank (PDB) is the main repository of biomolecular structure data. Lets see what is in it:

```
stats <- read.csv("pdb_stats.csv", row.names = 1)
head(stats)
```

```
                         X.ray    EM   NMR Integrative Multiple.methods Neutron
Protein (only)          178795 21825 12773         343               226      84
Protein/Oligosaccharide  10363  3564    34           8                11       1
Protein/NA                9106  6335   287          24                 7       0
Nucleic acid (only)       3132   221  1566           3                15       3
Other                      175    25    33           4                 0       0
Oligosaccharide (only)      11     0     6           0                 1       0
                        Other  Total
Protein (only)             32 214078
Protein/Oligosaccharide     0  13981
Protein/NA                  0  15759
Nucleic acid (only)         1   4941
Other                       0    237
Oligosaccharide (only)      4     22
```

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
n.sums <- colSums(stats)
n <- n.sums/n.sums["Total"]
round(n, digits = 2)
```

```
           X.ray               EM              NMR      Integrative
            0.81             0.13             0.06             0.00
Multiple.methods          Neutron            Other            Total
            0.00             0.00             0.00             1.00
```

94 percent of structures are solved by X-ray and Electron Microscopy

What is the total number of entries in the PDB?

there are 2.49018×10^5 structures in the PDB

```
n.sums["Total"]
```

```
 Total
249018
```

Q2: What proportion of structures in the PDB are protein?

```
sum(stats[,1:3])/sum(stats["Total"])
```

```
[1] 0.9969199
```

0.9969199 percent of structures in the PDB are protein.

Using Molstar

We can use the main Molstar viewer online

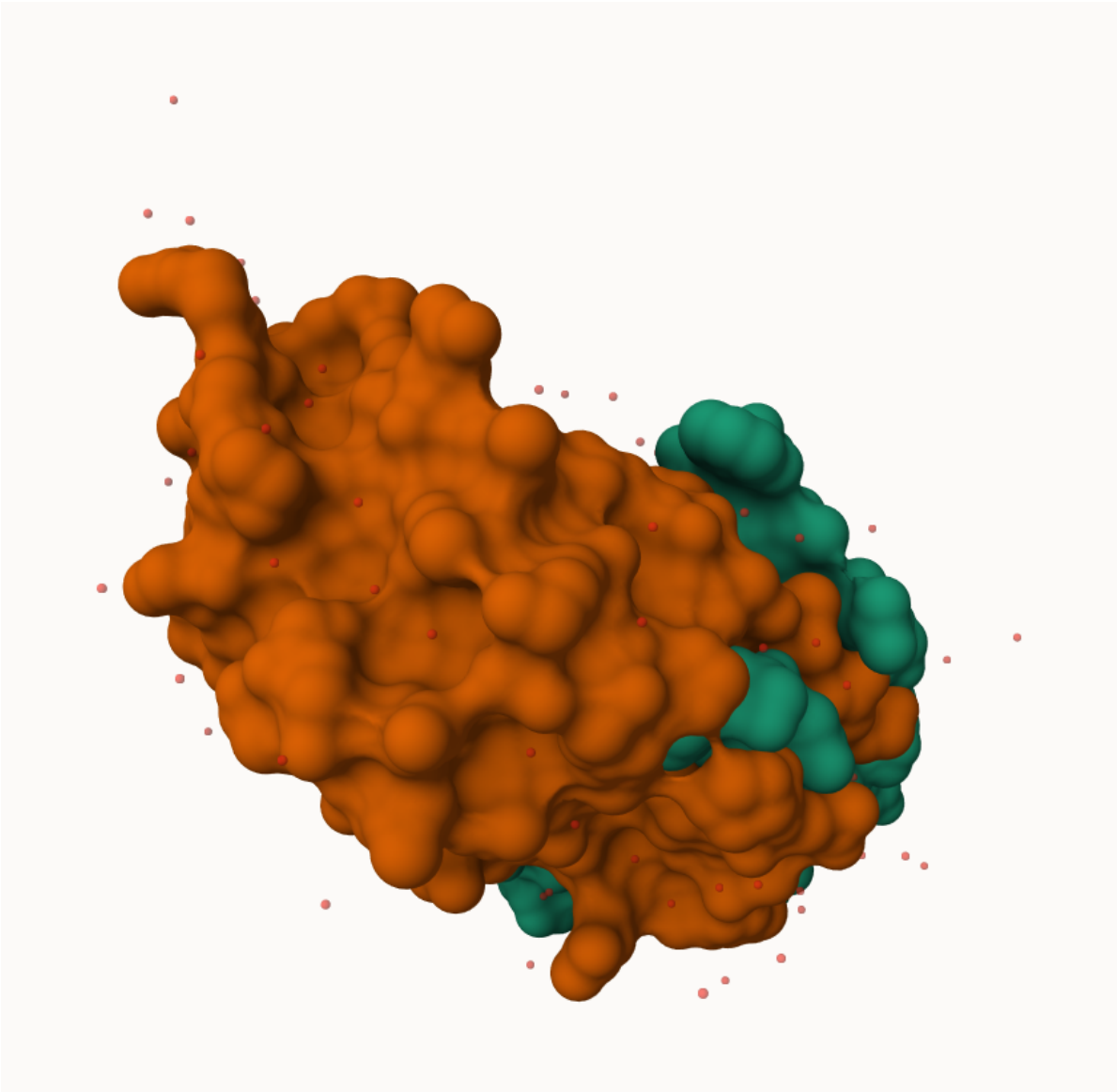
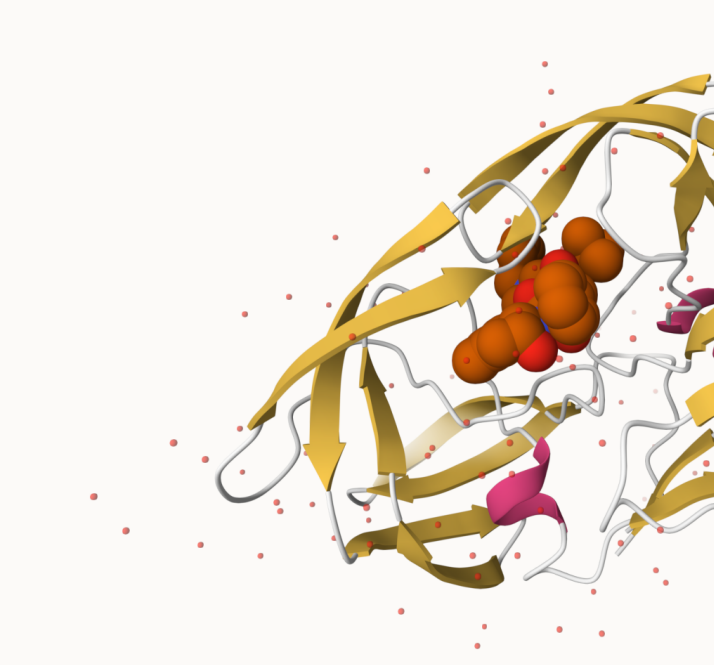


Figure 1: First view of HIV-Pr dimer with bound inhibitor

Q. Generate and insert an image of the HIV-Pr cartoon colored by secondary struc-



ture, showing the inhibitor (ligand) in spacefilling.

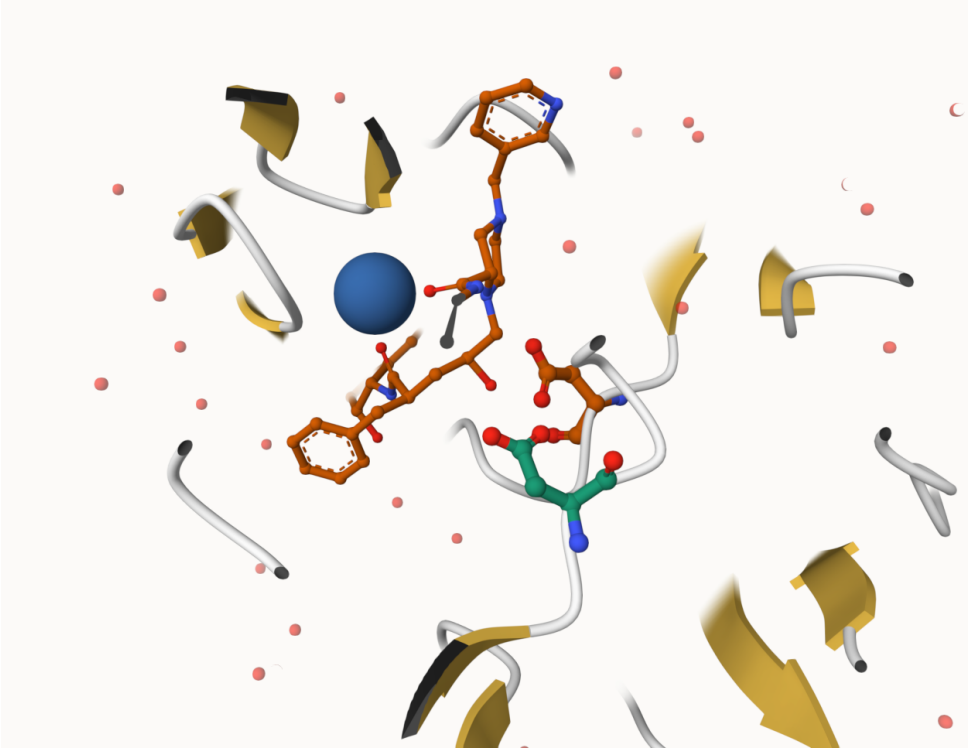
Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

The program set water molecules as a single point to remove clutter from the image. When 3 atoms are used the image becomes busier and harder to understand what is being shown.

Q5: There is a critical "conserved" water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

Yes, I can tell the water sits near a phenylalanine but I cannot see which residue number it is.

Q6. One final image showing catalytic APS 25 as ball and stick and the billion dollar



water molecule as space filling

The Bio3D package for structual bioinformatics

```
library(bio3d)

hiv <- read.pdb("1HSG")
```

```
  Note: Accessing on-line PDB file
```

```
hiv
```

```
 Call:  read.pdb(file = "1HSG")

   Total Models#: 1
     Total Atoms#: 1686,  XYZs#: 5058  Chains#: 2  (values: A B)

     Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)
```

```
     Non-protein/nucleic Atoms#: 172  (residues: 128)
     Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]

   Protein sequence:
      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
      QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
      ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
      VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

```r
head(hiv$atom)
```

```
  type eleno elety  alt resid chain resno insert      x      y     z o     b
1 ATOM     1     N <NA>   PRO     A     1   <NA> 29.361 39.686 5.862 1 38.10
2 ATOM     2    CA <NA>   PRO     A     1   <NA> 30.307 38.663 5.319 1 40.62
3 ATOM     3     C <NA>   PRO     A     1   <NA> 29.760 38.071 4.022 1 42.64
4 ATOM     4     O <NA>   PRO     A     1   <NA> 28.600 38.302 3.676 1 43.40
5 ATOM     5    CB <NA>   PRO     A     1   <NA> 30.508 37.541 6.342 1 37.87
6 ATOM     6    CG <NA>   PRO     A     1   <NA> 29.296 37.591 7.162 1 38.40
  segid elesy charge
1  <NA>     N   <NA>
2  <NA>     C   <NA>
3  <NA>     C   <NA>
4  <NA>     O   <NA>
5  <NA>     C   <NA>
6  <NA>     C   <NA>
```

```r
pdbseq(hiv)
```

```
  1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18  19  20
"P" "Q" "I" "T" "L" "W" "Q" "R" "P" "L" "V" "T" "I" "K" "I" "G" "G" "Q" "L" "K"
 21  22  23  24  25  26  27  28  29  30  31  32  33  34  35  36  37  38  39  40
"E" "A" "L" "L" "D" "T" "G" "A" "D" "D" "T" "V" "L" "E" "E" "M" "S" "L" "P" "G"
 41  42  43  44  45  46  47  48  49  50  51  52  53  54  55  56  57  58  59  60
"R" "W" "K" "P" "K" "M" "I" "G" "G" "I" "G" "G" "F" "I" "K" "V" "R" "Q" "Y" "D"
 61  62  63  64  65  66  67  68  69  70  71  72  73  74  75  76  77  78  79  80
"Q" "I" "L" "I" "E" "I" "C" "G" "H" "K" "A" "I" "G" "T" "V" "L" "V" "G" "P" "T"
 81  82  83  84  85  86  87  88  89  90  91  92  93  94  95  96  97  98  99   1
```

```
"P" "V" "N" "I" "I" "G" "R" "N" "L" "L" "T" "Q" "I" "G" "C" "T" "L" "N" "F" "P"
  2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18  19  20  21
"Q" "I" "T" "L" "W" "Q" "R" "P" "L" "V" "T" "I" "K" "I" "G" "G" "Q" "L" "K" "E"
 22  23  24  25  26  27  28  29  30  31  32  33  34  35  36  37  38  39  40  41
"A" "L" "L" "D" "T" "G" "A" "D" "D" "T" "V" "L" "E" "E" "M" "S" "L" "P" "G" "R"
 42  43  44  45  46  47  48  49  50  51  52  53  54  55  56  57  58  59  60  61
"W" "K" "P" "K" "M" "I" "G" "G" "I" "G" "G" "F" "I" "K" "V" "R" "Q" "Y" "D" "Q"
 62  63  64  65  66  67  68  69  70  71  72  73  74  75  76  77  78  79  80  81
"I" "L" "I" "E" "I" "C" "G" "H" "K" "A" "I" "G" "T" "V" "L" "V" "G" "P" "T" "P"
 82  83  84  85  86  87  88  89  90  91  92  93  94  95  96  97  98  99
"V" "N" "I" "I" "G" "R" "N" "L" "L" "T" "Q" "I" "G" "C" "T" "L" "N" "F"
```

Let's try out the new **bio3dview** package that is not yet on CRAN. We can use the **remotes** package to install any R package from GitHub.

Quick viewing of PDBs

```r
library(bio3dview)

sele <- atom.select(hiv, resno=25)

#view.pdb(hiv, backgroundColor = "lightblue",
#         highlight = sele,
#         highlight.style = "spacefill")
```

Prediction of Protein Flexibility

```r
#perform flexibility prediction
adk <- read.pdb("6s36") #reads Adenylate Kinase
```

```
  Note: Accessing on-line PDB file
   PDB has ALT records, taking A only, rm.alt=TRUE
```

```r
adk
```

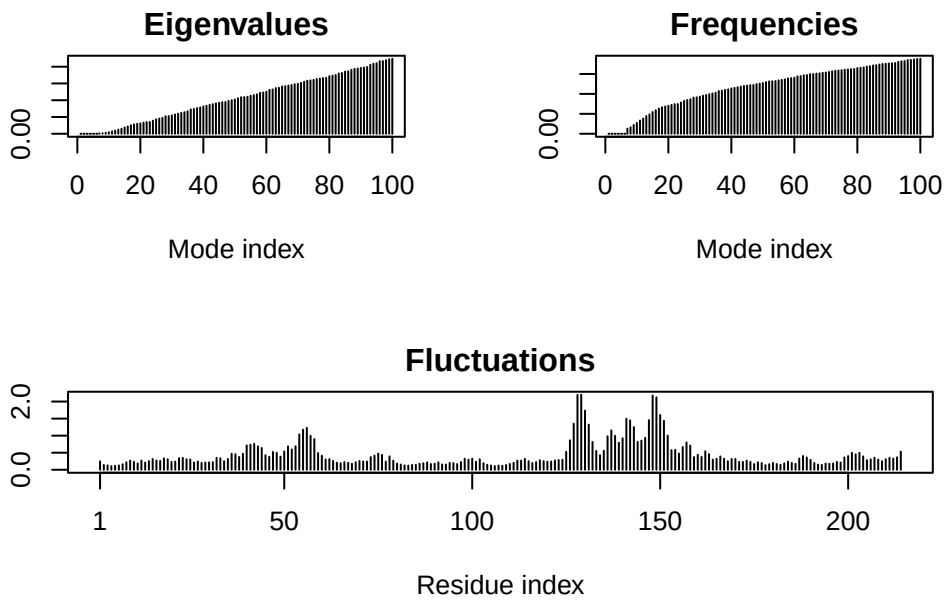
```
 Call:  read.pdb(file = "6s36")
```

```
   Total Models#: 1
     Total Atoms#: 1898,  XYZs#: 5694  Chains#: 1  (values: A)

     Protein Atoms#: 1654  (residues/Calpha atoms#: 214)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 244  (residues: 244)
     Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]

   Protein sequence:
      MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
      DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI
      VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
      YYSKEAEAGNTKYAKVDGTKPVAEVRADLEKILG

+ attr: atom, xyz, seqres, helix, sheet,
        calpha, remark, call
```

```r
m <- nma (adk) #mathmatically calculates the flexibility
```

```
 Building Hessian...        Done in 0.052 seconds.
 Diagonalizing Hessian...   Done in 0.608 seconds.
```

```r
plot(m) #plots the flexibility calculations
```



Write our results as a wee trajectory:

```
mktrj(m, file = "results.pdb") # wrote out the results
```

```
#view.nma(m)
#view.nma()
```

Comparitive Analysis